WHERE AUTONOMY MEETS PRACTICAL DELIVERY

As part of IEEE SMC 2026, held October 4–7, 2026, in Bellevue, WA, USA, the AICA Challenge provides students with a hands-on opportunity to develop innovative AI solutions using Quanser Interactive Labs (QLabs), the QCar2 ground vehicle, and the QDrone2 aerial vehicle platforms. The challenge is based on a realistic logistics scenario where packages are delivered from a pickup location to customers through path planning, control algorithms, and AI-driven decision-making.

COMPETITION HIGHLIGHTS

- Hands-on simulation environment in Quanser Interactive Labs (QLabs).
- Develop and test strategies using Python or MATLAB/Simulink.
- Practical delivery tasks with the QCar2 ground vehicle and QDrone2 aerial vehicle.
- Planning, coordination, and control algorithms for effective mission execution.

Registration Open Now

Register your team and begin preparing for the AICA Challenge 2026.

2026 Challenge Theme: Collaborative Autonomy in Simulation

Teams will develop and test AI-driven coordination strategies in Quanser Interactive Labs (QLabs), using the QCar2 ground vehicle and QDrone2 aerial vehicle platforms.

The mission focuses on a realistic logistics scenario where autonomous platforms coordinate to deliver packages from a central pickup location to customers.

Why Participate

- **$2,000 first-place prize** for the winning team.
- Visibility within the IEEE SMC community.
- Hands-on experience with next-generation autonomous systems.
- Opportunity to showcase innovative AI-driven solutions.

COMPETITION STAGES

STAGE 1

Virtual Stage

Video Submission

Teams develop and test their solutions in Quanser Interactive Labs (QLabs) using the provided Virtual Stage scenario, then submit a video link demonstrating their strategy and mission execution.

STAGE 2

On-Site Finals

October 4–7, 2026

Top teams from the Virtual Stage will be invited to compete at IEEE SMC 2026, where they will execute and demonstrate solutions based on the On-Site Finals scenario.

REGISTER TODAY

For the AICA Challenge 2026, visit the official home portal for full competition information and registration details.

AICA Challenge Home Portal

Contact: smc.aica.2026@gmail.com